

Theorem 1. Suppose that $\{x_n\}$ and $\{y_n\}$ are real sequences. If $\{x_n\}$ and $\{y_n\}$ are convergent, then

$$\lim_{n \rightarrow \infty} (x_n + y_n) = \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n.$$

Proof. Let $L = \lim_{n \rightarrow \infty} x_n$ and $M = \lim_{n \rightarrow \infty} y_n$. Given $\varepsilon > 0$, there exist $N_1, N_2 \in \mathbb{N}$ such that

$$n \geq N_1 \quad \text{implies} \quad |x_n - L| < \varepsilon/2$$

and

$$n \geq N_2 \quad \text{implies} \quad |y_n - M| < \varepsilon/2.$$

Take $N = \max\{N_1, N_2\}$, then $n \geq N$ implies

$$\begin{aligned} |(x_n + y_n) - (L + M)| &= |(x_n - L) + (y_n - M)| \\ &\leq |x_n - L| + |y_n - M| \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon. \end{aligned}$$

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